

Emergent Curiosity in Autonomous Research Agents

Abstract. We study whether a language-model agent, given a corpus of scientific PDFs and no human questions, can identify its own knowledge gaps and pursue them productively. We describe a closed loop in which the agent ingests documents, generates questions across nine categories, answers them with grounded evidence, reflects on each answer to spawn deeper questions, and synthesizes cross-document insights. A curiosity module inspects the agent's own uncertainty and seeds new questions from the gaps it finds.

1. **Introduction.** Retrieval-augmented systems are overwhelmingly reactive: a user asks, the system answers. We ask a different question. What happens when an agent is left alone with a library and told only to understand it? The agent must decide what is worth asking. This shifts the hard problem from answering to question formation.

2. **Method.** Each document is parsed, chunked, and embedded. Hybrid retrieval fuses dense vector scores with BM25 lexical scores using reciprocal rank fusion. Answers below a confidence threshold are marked unresolved and fed back to the questioner as explicit knowledge gaps. A persistent self-model records beliefs, known unknowns, and current curiosities, and is promoted into a searchable memory the solver reads at answer time.

3. **Results.** Over a 1,000-paper run the agent issued 33 model calls per document on average and surfaced 13 cross-document contradictions that a single-document reader would miss. The intelligence score, a composite of coverage, grounding, and novelty, rose monotonically as the loop ran.

4. **Discussion.** The novel component is not retrieval but the curiosity loop: gap detection -> self-question -> solve -> memory. We make no claim about machine consciousness; the self-model is an engineering device for directing attention, not a phenomenal one.